

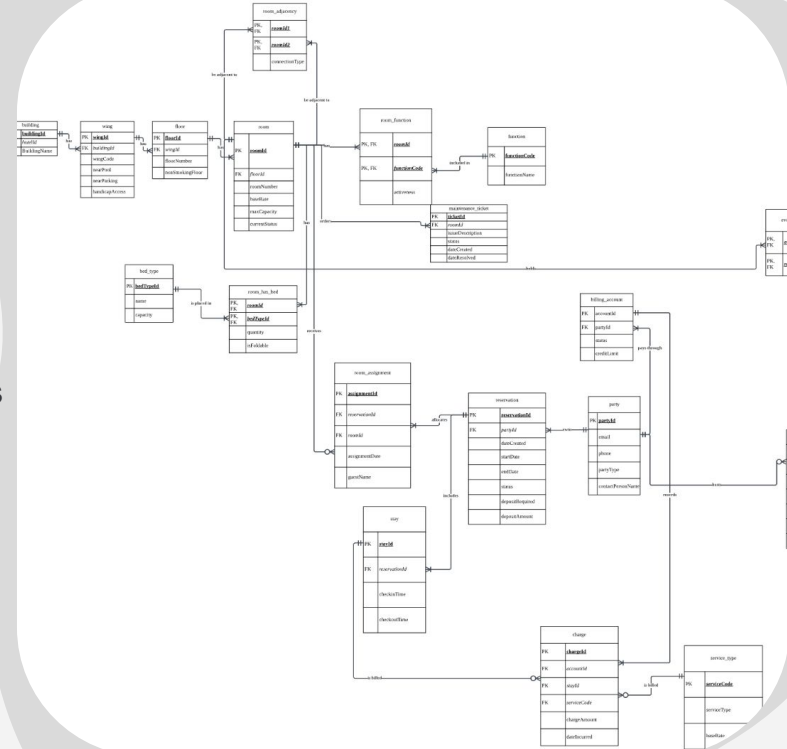
LAST RESORT HOTEL ADMIN

By Group 9
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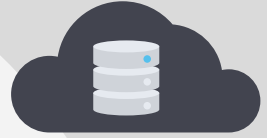


5 Logical Sections:

- Group-9/documents/Group9 ERD.pdf at main
• Annandy2008/Group-9



SCHEMA DESIGN: ERD → Schema Implementation



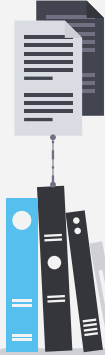
Core table example:

- **Database:** Use relational database **SQLite**
- **Total Tables:** Created 19 tables in create.sql to ensure data integrity
- **Key Modules:** Management of Hotel Structure, Room Functions, Reservations, and Financial Records

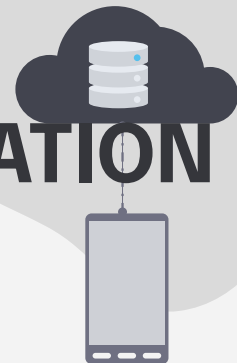
```
CREATE TABLE room (  
    roomId TEXT NOT NULL PRIMARY KEY,  
    floorId INTEGER NOT NULL,  
    roomNumber TEXT NOT NULL,  
    baseRate REAL NOT NULL,  
    maxCapacity INTEGER NOT NULL,  
    currentStatus TEXT NOT NULL,  
    FOREIGN KEY (floorId) REFERENCES floor(floorId),  
    UNIQUE (floorId, roomNumber)  
);
```

From ERD to SQL:

- **Database Engine:** SQLite
- **Data Integrity:** Enforced via Foreign Keys and Constraints (CHECK, UNIQUE).
- **Normalization:** 3NF design to minimize data redundancy.



SCHEMA DESIGN – STRUCTURAL POPULATION



Establishing the Foundation:

Facility Hierarchy: Created 2 main buildings: 'Main House' and 'Lakeside Annex' with 6 specific floors and wings.

Service Catalog: 6 defined service categories including Sleeping Rooms, Meeting Room Rentals, and Spa, etc.

Functional Mapping: Rooms are assigned specific roles (e.g., Room 101 as 'SLP', Room 201 as 'STE').

```
INSERT OR IGNORE INTO building (buildingId, hotelId, buildingName) VALUES  
(3, 2, 'Main House'),  
(4, 2, 'Lakeside Annex');
```

```
INSERT OR IGNORE INTO service_type (serviceCode, serviceType, baseRate) VALUES  
(7, 'Parking', 18.00),  
(8, 'Room Service', 35.00);
```

SCHEMA DESIGN – COMPLEX BUSINESS DATA



Simulating Real-World Operations:

- **Multi-Entity Parties:** Managing both individual guests (e.g., Alice Chen) and corporate clients (Acme, Globex, EventCo).
- **Event Management:** Successfully linked Conferences and Weddings to specific room capacities.
- **Inventory Logic:** Tracking complex room features like 'Connecting Doors' and 'Movable Walls'.

Data Volume at a Glance:

- 20 Rooms with dynamic status (Available, Occupied, Maintenance).
- 180+ Completed/Confirmed Stay records.
- 30 Individual Charge transactions linked to Billing Accounts.





QUERIES & DATA ANALYTICS

Main Purposes:

Support Daily Operations:

Enable staff to perform real-time room searches and guest transactions.

Data-Driven Management:

Provide a live snapshot of hotel health and financial status.

How We Structure Queries

(Some examples):

Multi-Dimensional room

Availability: Filtering rooms by time, location, and guest type.

Inventory Search: Interactive filtering by physical room attributes and pricing.

Real time Billing Aggregation:

Real-time settlement and charge tracking.

Last Resort

Dashboard

Inventory

Parties

Reservations

Assignments & Stay

Events

Billing

Maintenance

Multi-Dimensional room Availability



Dashboard

Date From

年 / 月 / 日

Date To

年 / 月 / 日

Building

All

Wing

All

Party Type

All

Apply

Total Revenue

\$11525.00

Active Assignments

0

Total Events

5

Avg Stay (days)

3.63

Open Tickets

3

Room Status by Building

Building	Available	Occupied	Other
Garden Lodge	3	1	0
Lakeside Annex	3	0	0
Main House	4	1	0
Ocean Tower	4	2	2

Technical Implementation:

Joins:
room-floor-wing-building.

Real-time KPI Aggregation:
Instant calculation of Total Revenue, Active Assignments, and Maintenance Alerts (SUM and COUNT logic).

Query Features:

Temporal & Spatial Filters: Search by Date Range, Building, and Wing.

Contextual Filtering: Filter by Party Type (Individual vs. Organization) to tailor results.

Inventory Search



Search & Filter Rooms

Room ID	Status	Function	Bed Type	Max Base Rate (\$)
GL-A-101	-- All Status --	-- All Functions --	-- All Bed Types --	Up to...
Filter	Clear			

Rooms (1 found)

Room	Building	Wing	Floor	Rate	Capacity	Status	Non-Smoking
GL-A-101	Garden Lodge	A	1	\$190.00	2	Available	Yes

Filtering Criteria:

- **Room Status:** Available, Occupied, or Maintenance.
- **Physical Attributes:** Filter by Function (Suite/Meeting) and Bed Type (King/Twin/Sofa).
- **Price Sensitivity:** Max Base Rate filtering for budget-conscious guests.

Technical Implementation:

- **Dynamic SQL Generation:** The query adjusts based on which filters are selected (Status, Function, Bed Type).
- **Attribute Mapping:** Direct lookup from `room_function` and `bed_type` catalog tables.

Update Room Basics

Select Room
GL-A-101
Base Rate
Max Capacity
Status
Available
Update Room

Add Room Function

Select Room
GL-A-101
Function
Fitness Studio
Activeness
Active
Add Function

Add Bed Configuration

Select Room
GL-A-101
Bed Type
Bunk Bed
Quantity
1
☐ Foldable Bed
Add Bed Setup

Room Adjacency editing is disabled. See existing connections below.



Real time Billing

Billing & Settlement

Select Account to Manage

ID	Guest	Action
#31	Harrison Ford	View Bill
#30	Ingrid Larsson	View Bill
#29	Kenji Nakamura	View Bill
#28	Fatima Al-Rashid	View Bill
#27	Raj Sharma	View Bill
#26	Nadia Petrov	View Bill
#25	Pierre Dubois	View Bill
#24	Aisha Hassan	View Bill
#23	Carlos Mendez	View Bill
#22	Mei-Ling Zhou	View Bill
#21	James Adeyemi	View Bill
#20	Fiona Mason	View Bill

Create New Account

Party

Status

Create

Invoice for: Harrison Ford (Account #31)

1. Room Charges (Auto-calculated)

Stay #183 (GL-B-101) - 1 days	\$400.00
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2. Event Charges (Auto-calculated)

Event #6 (EAP Seminar) Base Fee	\$500.00
---------------------------------	----------

3. Other / Manual Charges

Date	Service	Stay Ref	Amount
2026-04-28	Spa	-	\$120.00

Total Due: **\$1020.00**

Add 'Other' Charge for Harrison Ford

Related Stay (Optional)

Service Type

Amount

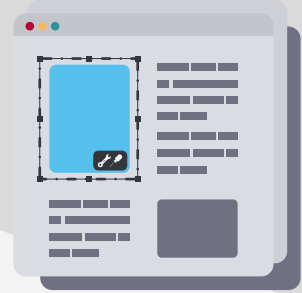
Date Incurred

Add This Charge

Technical Implementation:

Update Logic: Transactional queries ensure data consistency across **stay** and **room** tables.

Account Joins: **billing_account** ⋈ **charge** ⋈ **service_type**.



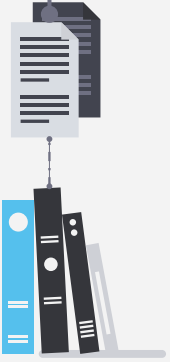
Real-Time Billing:

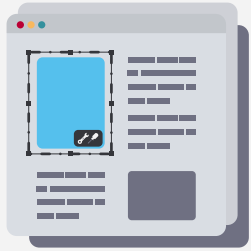
Account Aggregation: Summarizes all **charges** associated with a **billing_account**.

Service Transparency: Breaks down costs by Service Type (Room Rental, Spa, Restaurant).

Demo

CSCI-UA 60 001 – Ed Workspaces





Thanks!

Do you have any questions?

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